

The background features a dark purple gradient with large, overlapping, semi-transparent shapes in shades of orange, pink, and blue, creating a dynamic, abstract design.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

SUP307-R

Improve operation excellence for production workloads

Tipu Qureshi

Sr. Principal Support Engineer
AWS

Anish Sheth

Sr. Solutions Architect
AWS



Agenda

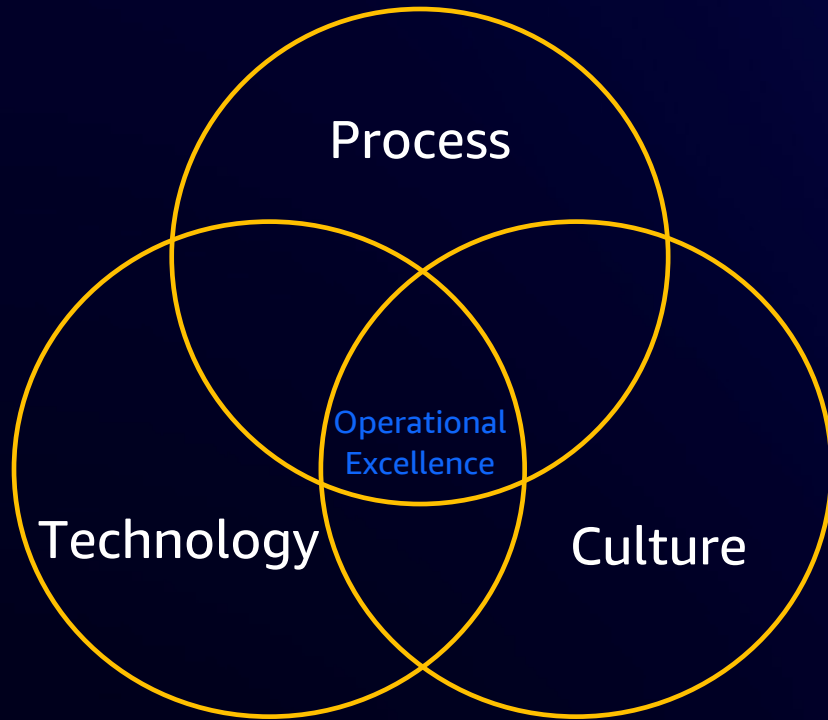
1. What is operational excellence?
2. How can I achieve operational excellence?
3. AWS services and best practices to assist your journey
4. Q&A/demo

“Everything fails, all the time.”

Dr. Werner Vogels
VP and CTO at Amazon.com



What is operational excellence?



- Defining operational goals
- Detecting and addressing operational issues
- Effectively responding to operational issues
- Exceeding operational goals
- Continually improving operations
- Delighting customers

...achieving all this at cloud scale

Scale brings challenges and opportunities



What is the AWS Well-Architected Framework?



Pillars and lenses



Design principles



Questions



Best practices

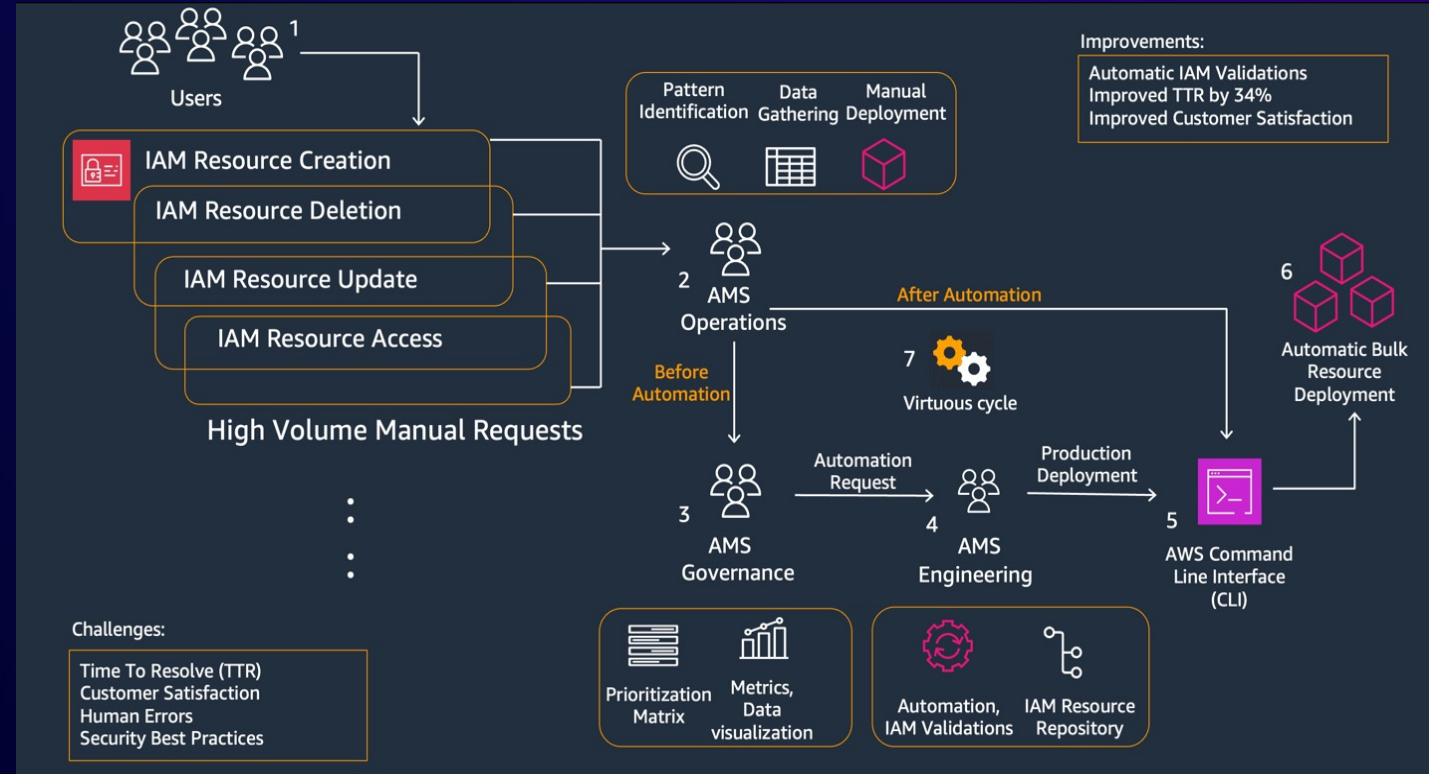
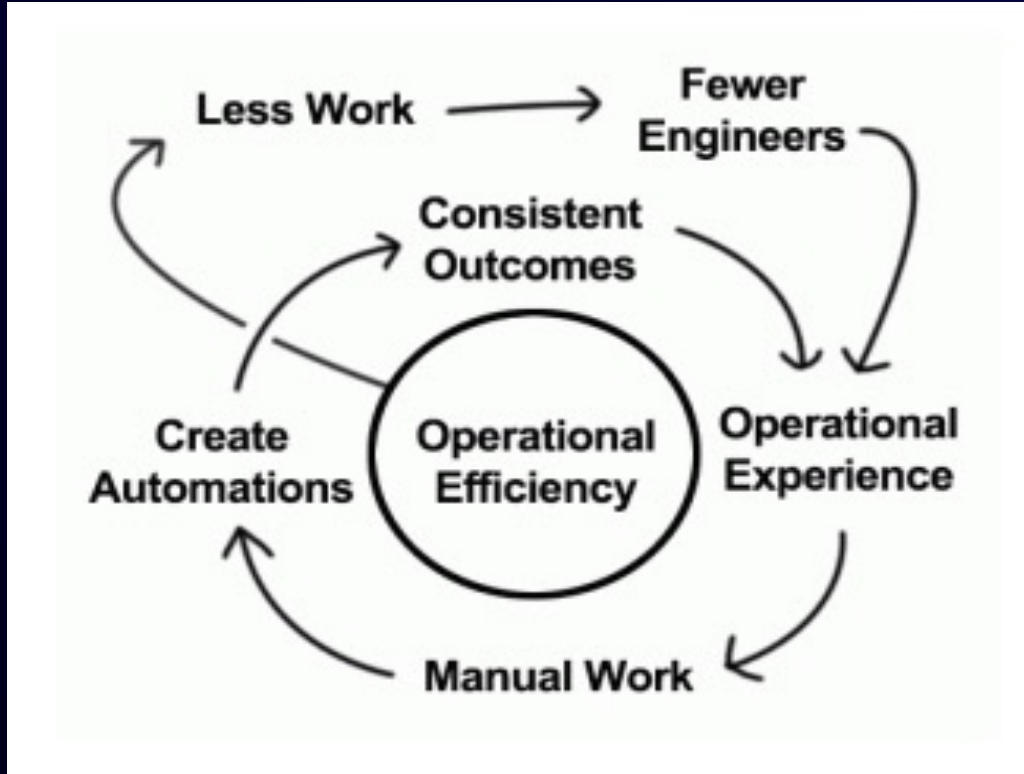
Operational excellence design principles

- Perform operations as code
- Make frequent, small, reversible changes
- Refine operations procedures frequently
- Anticipate failure
- Learn from all operations failures



Operational
excellence

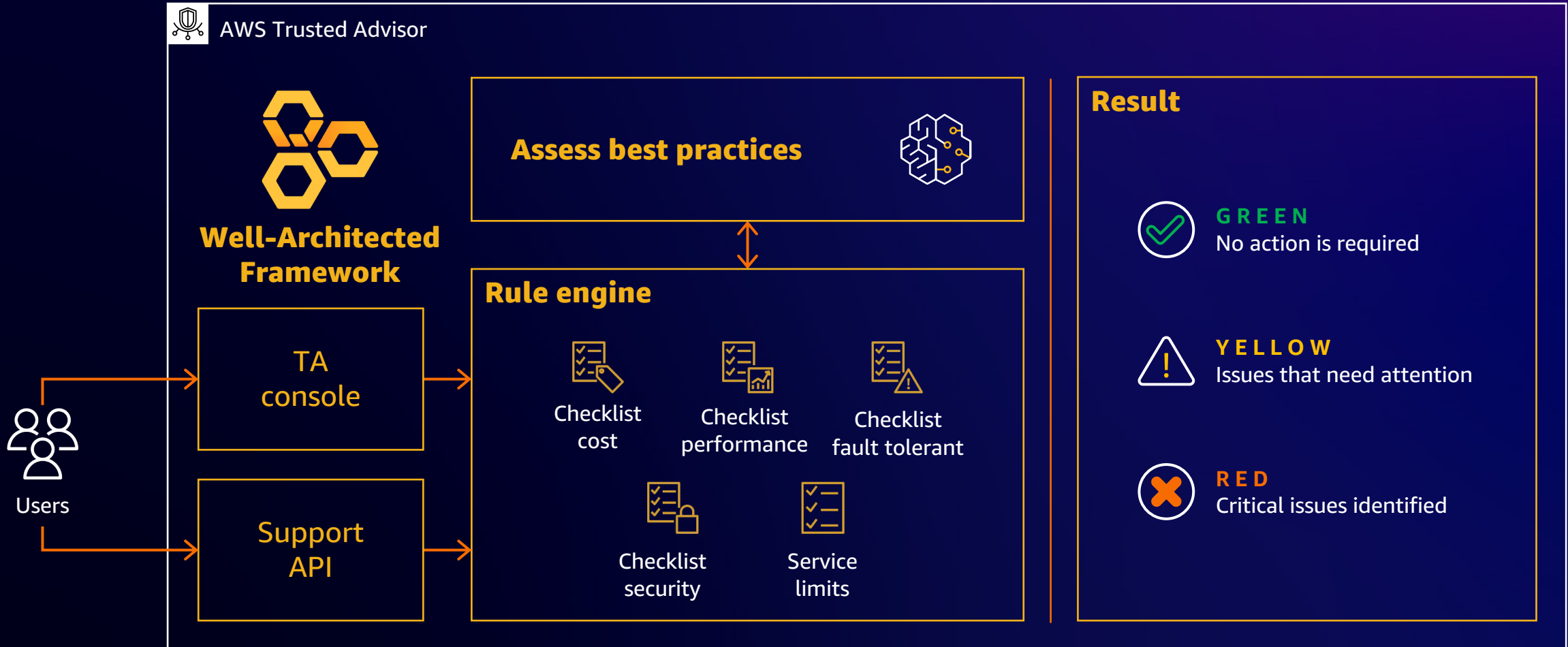
Operations Flywheel fueling automation



Operational excellence driven by automation and AI

AWS Trusted Advisor

RECOMMENDATIONS TO HELP YOU FOLLOW BEST PRACTICES



Intelligent data signals to automations



Amazon CloudWatch
anomaly detection



AWS Trusted Advisor



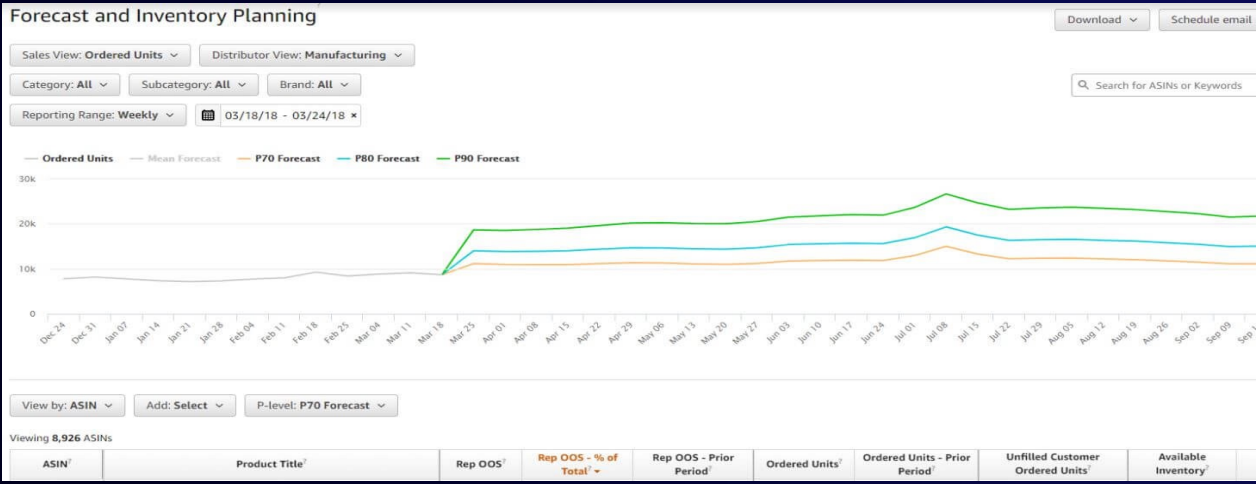
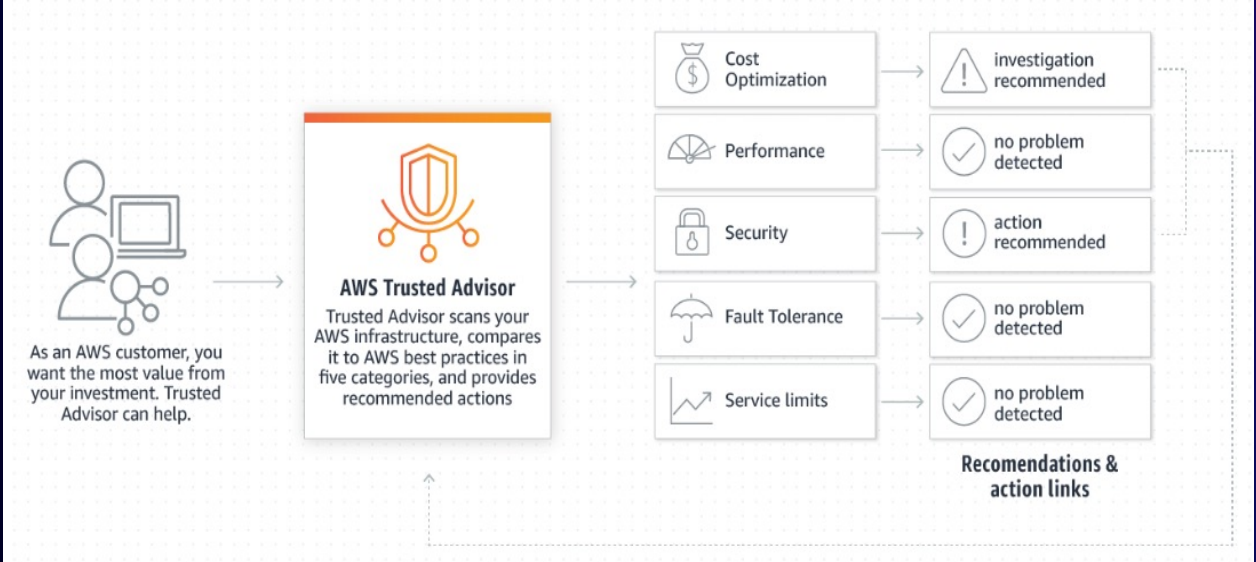
AWS Health



Amazon Forecast



Amazon DevOps Guru



Amazon EventBridge



Amazon
Comprehend



AWS Systems
Manager



AWS Lambda



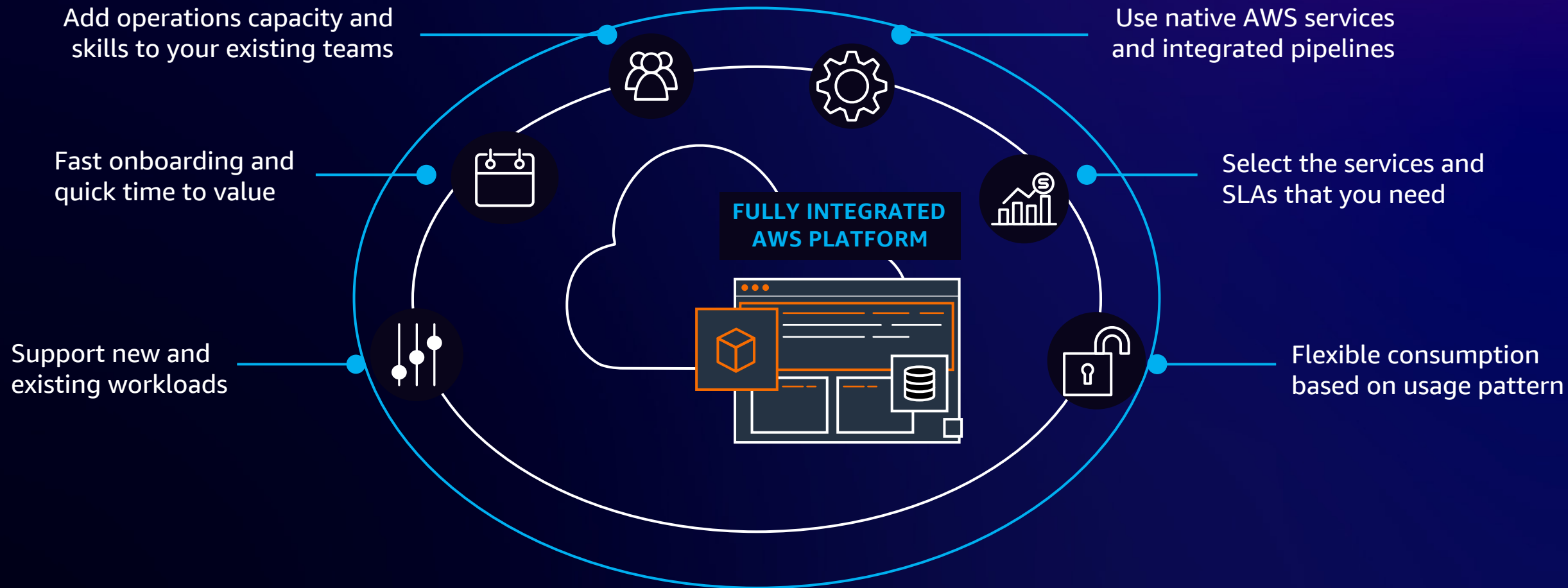
AWS Support for operational excellence



AWS support plans for every stage of your cloud journey



AWS Managed Services (AMS)



AWS Incident Detection and Response

24/7 PROACTIVE MONITORING AND CRITICAL INCIDENT MANAGEMENT FOR SELECTED WORKLOADS

Improved observability

Define critical alarms on the application layer

Observability on AWS infrastructure and services

Early incident detection

24/7 workload monitoring

AWS engages you within 15 minutes of an incident

Rapid help from AWS incident management engineer

Faster resolution

Access to experts to guide you through recovery

Predefined response plans and runbooks

Live engagement

Reduced potential for failure

Continuous learning mechanism

Proactive services

Post-incident review

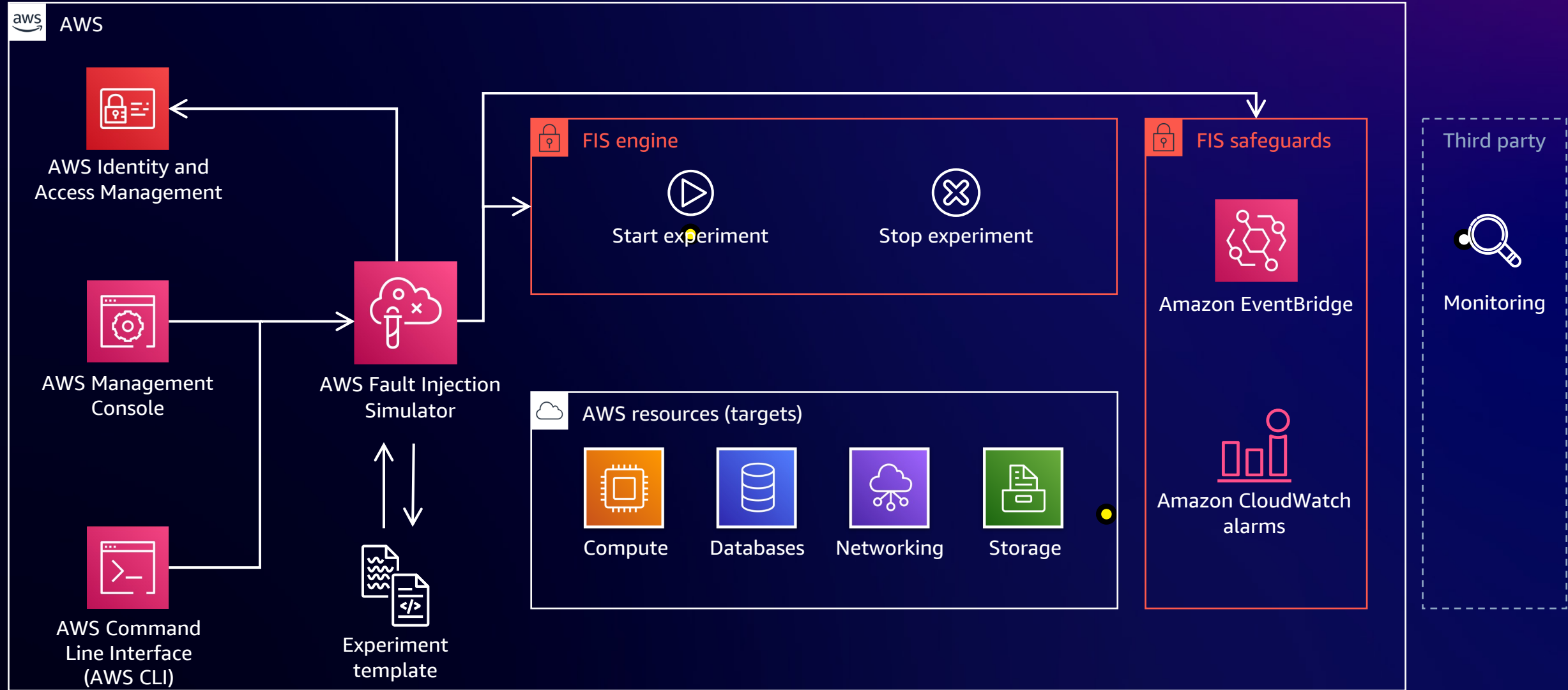
Demo



Improve operational excellence



AWS Fault Injection Service – Chaos engineering



Incident Manager, a capability of AWS Systems Manager

1 Prepare response plans

Incident Manager response plan



Escalation plans and contact

Who you engage



Automated runbooks

What you do



AWS Chatbot

Where you collaborate

2 Monitor and detect issues

Triggers (event sources) and 3rd-party integrations with EventBridge



CloudWatch alarms



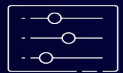
EventBridge rules



Manual detection/3rd-party
monitoring systems

3 Respond and resolve

Incident Manager dashboard



Incident timeline



Integrated runbook



Triage and analysis

Post Incident Analysis Correction of Errors (COE)

A **mechanism** to:

- Identify and fix problems

- Drive ownership of action items

- Maximize lessons learned

- Prevent recurrence of the problem

Not to blame

Not to punish

Thank you!

Tipu Qureshi

tqquresh@amazon.com

Anish Sheth

anisheth@amazon.com



Please complete the session
survey in the mobile app



Demo architecture

